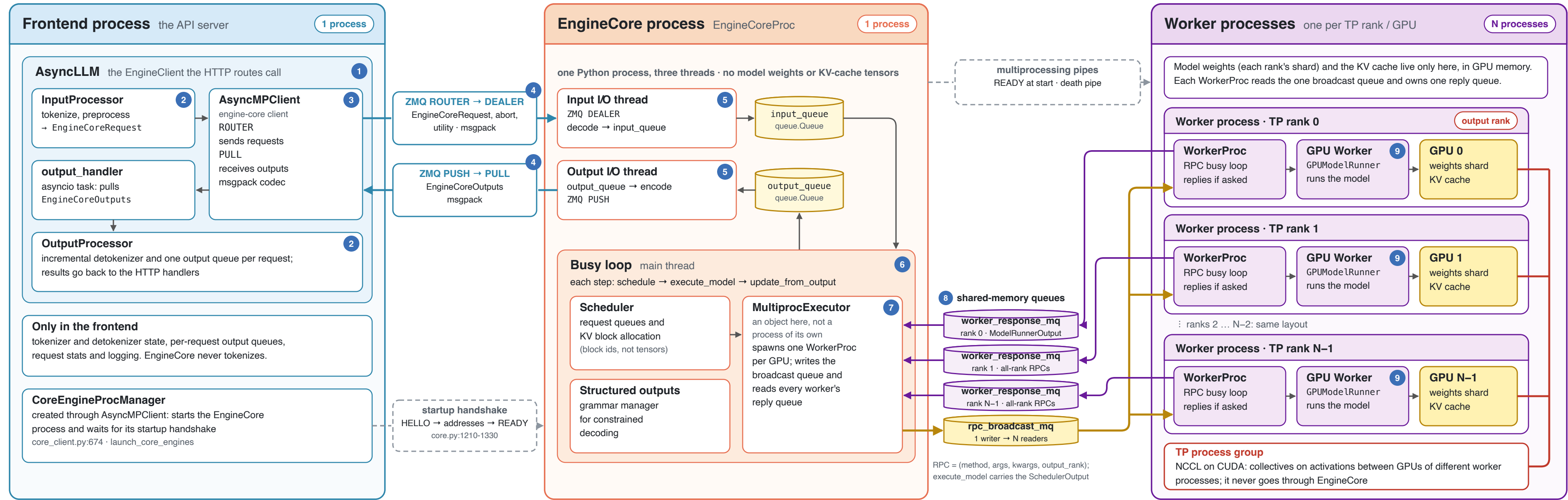


vLLM V1 on one node: which OS processes exist and what crosses between them

Tensor parallelism N with the multiprocess executor (DP = 1, PP = 1). Structure, not step order.

3 process roles · N + 2 OS processes

ONE HOST



Code anchors badge numbers are the answer-key facts; paths under vllm/v1/

[1] engine/async_llm.py:80
[2] async_llm.py:157, 167 · input_processor.py:41 · output_processor.py:51
[3] async_llm.py:180 · core_client.py:137-175, 1086
[4] core_client.py:616-662 · core.py:1775, 1878
[5] core.py:1088-1199, 1473, 1757-1894

[6] core.py:140, 152, 168, 630-660
[7] executor/multiproc_executor.py:111, 195
[8] multiproc_executor.py:164-170, 619, 1032-1036
[9] worker/gpu_worker.py:536
[+] TP group gpu_worker.py:491, 1586-1624 · output rank multiproc_executor.py:543-557

→ ZMQ socket → broadcast queue (shared memory, 1 → N) → reply queue owned by one worker → NCCL collective → startup / lifecycle only → in-process call or queue

vLLM @ 4c2d277